KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3301 | 3 Credits: 3 | Program Core | Source-grounded

Problem Solving Using Python

● To learn and understand the basics of Python programming.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3301 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3301.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Robotic Process Automation
Course code	3301
Course title	Problem Solving Using Python
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L: 3 T: 0 P: 0) Periods per semester: 45
Periods per semester	45

Item	Official information
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

27 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● To learn and understand the basics of Python programming.
- ● Develop different programming applications using Python.
- ● To learn the concept of file handling and exception handling.

Course outcomes

CO	Official outcome	Study level
CO1	construct Python programs using control statements 11 Applying and expressions. Develop programs using functions and slicing	See official syllabus
CO2	14 Applying operation. Use Python data structures - List, Tuple, and	See official syllabus
CO3	9 Applying Dictionary - for solving problems. Explain files, exceptions, and modules in Python for	See official syllabus
CO4	9 Applying solving problems	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 3 3
CO2	CO2 3 3 3
CO3	CO3 3 3 3
CO4	CO4 3 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	programs using control statements and expressions.	5	As specified
Module 2	: Develop programs using functions and slicing operation.	7	As specified
Module 3	:	9	As specified
Module 4	: Explain files, exceptions, and modules in Python for solving problems	6	As specified

MODULE 1

programs using control statements and expressions.

This module is presented from the official syllabus scope for Problem Solving Using Python. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: programs using control statements and expressions.

- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

1 Understanding. algorithm, flow chart, and pseudo code. Illustrate the basic concept of a Python- identifiers,

1 Understanding. algorithm, flow chart, and pseudo code. Illustrate the basic concept of a Python- identifiers, is an official topic in Module 1 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. algorithm, flow chart, and pseudo code. Illustrate the basic concept of a Python- identifiers, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. algorithm, flow chart, and pseudo code. illustrate the basic concept of a python- identifiers, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding. algorithm, flow chart, and pseudo code. Illustrate the basic concept of a Python- identifiers, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 1 Understanding. algorithm, flow chart, and pseudo code. Illustrate the basic concept of a Python- identifiers, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

2 Understanding keywords, variables and data types. Explain operators and expressions- Expressions, Arithmetic, comparison, assignment, logical, bitwise,

2 Understanding keywords, variables and data types. Explain operators and expressions- Expressions, Arithmetic, comparison, assignment, logical, bitwise, is an official topic in Module 1 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding keywords, variables and data types. Explain operators and expressions- Expressions, Arithmetic, comparison, assignment, logical, bitwise, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding keywords, variables and data types. explain operators and expressions- expressions, arithmetic, comparison, assignment, logical, bitwise, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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3 Understanding membership and identity operators, precedence of operators, associativity and input output functions. Explain Control statements- if elif else statement,

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Study points

- Define or identify 3 Understanding membership and identity operators, precedence of operators, associativity and input output functions. Explain Control statements- if elif else statement, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding membership and identity operators, precedence of operators, associativity and input output functions. explain control statements- if elif else statement, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding membership and identity operators, precedence of operators, associativity and input output functions. Explain Control statements- if elif else statement, means in the official course context
Study lens	Principle → components or stages → result → application
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while statement, for statement, break and continue 3 Understanding statements.

while statement, for statement, break and continue 3 Understanding statements. is an official topic in Module 1 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify while statement, for statement, break and continue 3 Understanding statements. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
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Engineering or professional relevance

In Polytechnic practice, while statement, for statement, break and continue 3 understanding statements. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What while statement, for statement, break and continue 3 Understanding statements. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 1-[illegible] while statement, for statement, break and continue 3 Understanding statements. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Develop programs using Control statements. 2 Applying Problem Analysis: - Methodology of problem solving- Algorithm and flowchart with examples, pseudo code. Introduction to Python: - Identifiers, keywords, variables, data types. Operators, expressions, precedence of operators, associativity. Input and output functions. Control statements: if elif else statement, iteration statements- while, for. Break and continue statements. Programming examples.

Develop programs using Control statements. 2 Applying Problem Analysis: - Methodology of problem solving- Algorithm and flowchart with examples, pseudo code. Introduction to Python: - Identifiers, keywords, variables, data types. Operators, expressions, precedence of operators, associativity. Input and output functions. Control statements: if elif else statement, iteration statements- while, for. Break and continue statements. Programming examples. is an official topic in Module 1 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop programs using Control statements. 2 Applying Problem Analysis: - Methodology of problem solving- Algorithm and flowchart with examples, pseudo code. Introduction to Python: - Identifiers, keywords, variables, data types. Operators, expressions, precedence of operators, associativity. Input and output functions. Control statements: if elif else statement, iteration statements- while, for. Break and continue statements. Programming examples. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
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Engineering or professional relevance

In Polytechnic practice, develop programs using control statements. 2 applying problem analysis: - methodology of problem solving- algorithm and flowchart with examples, pseudo code. introduction to python: - identifiers, keywords, variables, data types. operators, expressions, precedence of operators, associativity. input and output functions. control statements: if elif else statement, iteration statements- while, for. break and continue statements. programming examples. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop programs using Control statements. 2 Applying Problem Analysis: - Methodology of problem solving- Algorithm and flowchart with examples, pseudo code. Introduction to Python: - Identifiers, keywords, variables, data types. Operators, expressions, precedence of operators, associativity. Input and output functions. Control statements: if elif else statement, iteration statements- while, for. Break and continue statements. Programming examples. means in the official course context
Study lens	Principle → components or stages → result → application
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Malayalam support: Module 1-□□ Develop programs using Control statements. 2 Applying Problem Analysis: - Methodology of problem solving- Algorithm and flowchart with examples, pseudo code. Introduction to Python: - Identifiers, keywords, variables, data types. Operators, expressions, precedence of operators, associativity. Input and output functions. Control statements: if elif else statement, iteration statements- while, for. Break and continue statements. Programming examples. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
programs using control statements and expressions.

            Explain the methodology of problem solving -
  M1.01                                                                 1
Understanding.
            algorithm, flow chart, and pseudo code.

            Illustrate the basic concept of a Python- identifiers,
  M1.02                                                                 2
Understanding
            keywords, variables and data types.
            Explain operators and expressions- Expressions,
            Arithmetic, comparison, assignment, logical, bitwise,
  M1.03                                                                 3
Understanding
            membership and identity operators, precedence of
            operators, associativity and input output functions.
            Explain Control statements- if elif else statement,
  M1.04     while statement, for statement, break and continue          3
Understanding
            statements.
  M1.05     Develop programs using Control statements.                  2
Applying


Contents:
Problem Analysis: -
Methodology of problem solving- Algorithm and flowchart with examples, pseudo
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

: Develop programs using functions and slicing operation.

This module is presented from the official syllabus scope for Problem Solving Using Python. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: : Develop programs using functions and slicing operation.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

built-in-functions, type conversion, type coercion, 2 Understanding. mathematical functions Explain the concept of user defined functions-

built-in-functions, type conversion, type coercion, 2 Understanding. mathematical functions Explain the concept of user defined functions- is an official topic in Module 2 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify built-in-functions, type conversion, type coercion, 2 Understanding. mathematical functions Explain the concept of user defined functions- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, built-in-functions, type conversion, type coercion, 2 understanding. mathematical functions explain the concept of user defined functions- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What built-in-functions, type conversion, type coercion, 2 Understanding. mathematical functions Explain the concept of user defined functions- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ built-in-functions, type conversion, type coercion, 2 Understanding. mathematical functions Explain the concept of user defined functions- □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

function definition, function calling, function 3 Understanding returning values Illustrate the difference between actual and formal

function definition, function calling, function 3 Understanding returning values Illustrate the difference between actual and formal is an official topic in Module 2 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify function definition, function calling, function 3 Understanding returning values Illustrate the difference between actual and formal using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, function definition, function calling, function 3 understanding returning values illustrate the difference between actual and formal should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What function definition, function calling, function 3 Understanding returning values Illustrate the difference between actual and formal means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

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parameters. Types of formal arguments- required, 2 Understanding. default and keyword arguments. Explain the concept of scope of variable-local and

parameters. Types of formal arguments- required, 2 Understanding. default and keyword arguments. Explain the concept of scope of variable-local and is an official topic in Module 2 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify parameters. Types of formal arguments- required, 2 Understanding. default and keyword arguments. Explain the concept of scope of variable-local and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
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Engineering or professional relevance

In Polytechnic practice, parameters. types of formal arguments- required, 2 understanding. default and keyword arguments. explain the concept of scope of variable-local and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What parameters. Types of formal arguments- required, 2 Understanding. default and keyword arguments. Explain the concept of scope of variable-local and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 2-□□ parameters. Types of formal arguments-required, 2 Understanding. default and keyword arguments. Explain the concept of scope of variable-local and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

1 Understanding. global scope and recursive function.

1 Understanding. global scope and recursive function. is an official topic in Module 2 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding. global scope and recursive function. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding. global scope and recursive function. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding. global scope and recursive function. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 1 Understanding. global scope and recursive function. □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Develop python programs using functions 2 Applying. Summarize the concept of string operations-

Develop python programs using functions 2 Applying. Summarize the concept of string operations- is an official topic in Module 2 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

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Study points

- Define or identify Develop python programs using functions 2 Applying. Summarize the concept of string operations- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop python programs using functions 2 applying. summarize the concept of string operations- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop python programs using functions 2 Applying. Summarize the concept of string operations- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Develop python programs using functions 2 Applying. Summarize the concept of string operations- [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

concatenation, repetition, membership and slicing 2 Understanding operations. Develop programs using the concept of string

concatenation, repetition, membership and slicing 2 Understanding operations. Develop programs using the concept of string is an official topic in Module 2 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify concatenation, repetition, membership and slicing 2 Understanding operations. Develop programs using the concept of string using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, concatenation, repetition, membership and slicing 2 understanding operations. develop programs using the concept of string should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What concatenation, repetition, membership and slicing 2 Understanding operations. Develop programs using the concept of string means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

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Malayalam support: Module 2-□□ concatenation, repetition, membership and slicing 2 Understanding operations. Develop programs using the concept of string □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

2 Applying operations. Functions: - Built-in-functions: -Type conversion, type coercion, Mathematical functions. User defined functions: - Function definition, function calling, function returning values. Arguments and parameters, required arguments, default arguments, keyword arguments. Recursive functions. Scope of variables-local and global scope. Programming examples. Strings: - Indexing, string operations- concatenation, repetition, membership and slicing. Built-in- functions. Programming examples. Use Python data structures – List, Tuple, and Dictionary – for solving

2 Applying operations. Functions: - Built-in-functions: -Type conversion, type coercion, Mathematical functions. User defined functions: - Function definition, function calling, function returning values. Arguments and parameters, required arguments, default arguments, keyword arguments. Recursive functions. Scope of variables-local and global scope. Programming examples. Strings: - Indexing, string operations- concatenation, repetition, membership and slicing. Built-in- functions. Programming examples. Use Python data structures – List, Tuple, and Dictionary – for solving is an official topic in Module 2 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying operations. Functions: - Built-in-functions: -Type conversion, type coercion, Mathematical functions. User defined functions: - Function definition, function calling, function returning values. Arguments and parameters, required arguments, default arguments, keyword arguments. Recursive functions. Scope of variables-local and global scope. Programming examples. Strings: - Indexing, string operations- concatenation, repetition, membership and slicing. Built-in- functions. Programming examples. Use Python data structures – List, Tuple, and Dictionary – for solving using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying operations. functions: - built-in-functions: -type conversion, type coercion, mathematical functions. user defined functions: - function definition, function calling, function returning values. arguments and parameters, required arguments, default arguments, keyword arguments. recursive functions. scope of variables-local and global scope. programming examples. strings: - indexing, string operations- concatenation, repetition, membership and slicing. built-in- functions. programming examples. use python data structures – list, tuple, and dictionary – for solving should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying operations. Functions: - Built-in-functions: -Type conversion, type coercion, Mathematical functions. User defined functions: - Function definition, function calling, function returning values. Arguments and parameters, required arguments, default arguments, keyword arguments. Recursive functions. Scope of variables-local and global scope. Programming examples. Strings: - Indexing, string operations- concatenation, repetition, membership and slicing. Built-in- functions. Programming examples. Use Python data structures – List, Tuple, and Dictionary – for solving means in the official course context
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Malayalam support: Module 2-□□ 2 Applying operations. Functions: - Built-in-functions: -Type conversion, type coercion, Mathematical functions. User defined functions: - Function definition, function calling, function returning values. Arguments and parameters, required arguments, default arguments, keyword arguments. Recursive functions. Scope of variables-local and global scope. Programming examples. Strings: - Indexing, string operations- concatenation, repetition, membership and slicing. Built-in- functions. Programming examples. Use Python data structures – List, Tuple, and Dictionary – for solving □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
:       Develop programs using functions and slicing operation.
            Illustrate the concept function, types of functions –
  M2.01     built-in-functions, type conversion, type coercion,        2
Understanding.
            mathematical functions
            Explain the concept of user defined functions-
  M2.02     function definition, function calling, function            3
Understanding
            returning values
            Illustrate the difference between actual and formal
  M2.03     parameters. Types of formal arguments- required,           2
Understanding.
            default and keyword arguments.

            Explain the concept of scope of variable-local and
  M2.04                                                                1
Understanding.
            global scope and recursive function.

  M2.05     Develop python programs using functions                    2
Applying.


            Summarize the concept of string operations-
  M2.06     concatenation, repetition, membership and slicing          2
Understanding
            operations.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

:

This module is presented from the official syllabus scope for Problem Solving Using Python. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: :
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

List and operations- concatenation, repetition, and in 1 Understanding operator Illustrate the concept of traversing a list and built-in-

List and operations- concatenation, repetition, and in 1 Understanding operator Illustrate the concept of traversing a list and built-in- is an official topic in Module 3 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List and operations- concatenation, repetition, and in 1 Understanding operator Illustrate the concept of traversing a list and built-in- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list and operations- concatenation, repetition, and in 1 understanding operator illustrate the concept of traversing a list and built-in- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List and operations- concatenation, repetition, and in 1 Understanding operator Illustrate the concept of traversing a list and built-in- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ List and operations- concatenation, repetition, and in 1 Understanding operator Illustrate the concept of traversing a list and built-in- □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

1 Understanding List methods-len(), append(), insert() etc.

1 Understanding List methods-len(), append(), insert() etc. is an official topic in Module 3 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding List methods-len(), append(), insert() etc. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding list methods-len(), append(), insert() etc. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding List methods-len(), append(), insert() etc. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 3-□□ 1 Understanding List methods-len(), append(), insert() etc. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□□.

Develop programs using List 1 Applying Explain the concept of Tuples- creation, accessing

Develop programs using List 1 Applying Explain the concept of Tuples- creation, accessing is an official topic in Module 3 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop programs using List 1 Applying Explain the concept of Tuples- creation, accessing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop programs using list 1 applying explain the concept of tuples- creation, accessing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop programs using List 1 Applying Explain the concept of Tuples- creation, accessing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 3-□□ Develop programs using List 1 Applying Explain the concept of Tuples- creation, accessing □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

values in Tuples and operations- concatenation, 1 Understanding repetition, and in operator Summarize built-in-Tuple functions- len(), max(),

values in Tuples and operations- concatenation, 1 Understanding repetition, and in operator Summarize built-in-Tuple functions- len(), max(), is an official topic in Module 3 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify values in Tuples and operations- concatenation, 1 Understanding repetition, and in operator Summarize built-in-Tuple functions- len(), max(), using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, values in tuples and operations- concatenation, 1 understanding repetition, and in operator summarize built-in-tuple functions- len(), max(), should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What values in Tuples and operations- concatenation, 1 Understanding repetition, and in operator Summarize built-in-Tuple functions- len(), max(), means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

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Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] values in Tuples and operations- concatenation, 1 Understanding repetition, and in operator Summarize built-in-Tuple functions- len(), max(), [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Understanding min() etc. – with examples.

1 Understanding min() etc. – with examples. is an official topic in Module 3 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding min() etc. – with examples. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding min() etc. – with examples. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding min() etc. – with examples. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 1 Understanding min() etc. – with examples. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Develop programs using Tuples 1 Applying Explain how dictionaries are created, accessing

Develop programs using Tuples 1 Applying Explain how dictionaries are created, accessing is an official topic in Module 3 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop programs using Tuples 1 Applying Explain how dictionaries are created, accessing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop programs using tuples 1 applying explain how dictionaries are created, accessing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop programs using Tuples 1 Applying Explain how dictionaries are created, accessing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 3-[illegible] Develop programs using Tuples 1 Applying Explain how dictionaries are created, accessing [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

1 Understanding values in dictionary and traversing operations Summarize built-in-dictionary functions with

1 Understanding values in dictionary and traversing operations Summarize built-in-dictionary functions with is an official topic in Module 3 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

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Study points

- Define or identify 1 Understanding values in dictionary and traversing operations Summarize built-in-dictionary functions with using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding values in dictionary and traversing operations summarize built-in-dictionary functions with should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding values in dictionary and traversing operations Summarize built-in-dictionary functions with means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 3-□□ 1 Understanding values in dictionary and traversing operations Summarize built-in-dictionary functions with □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

1 Understanding examples

1 Understanding examples is an official topic in Module 3 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding examples using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding examples should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding examples means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 3-□□ 1 Understanding examples □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Develop programs using Dictionary 1 Applying Lists: - Introduction: - values and accessing elements, List operations- concatenation, repetition and in operator. Traversing a List. Built-in List methods- len(), append(), insert(), index(), pop(), remove(), reverse() etc. Programming examples. Tuples: - Introduction: - creating Tuples, accessing values in Tuples, operations- concatenation, repetition and in operator. Built-in-Tuple functions- len(), max(), min() etc. Programming examples. Dictionaries:- Introduction: - creating Dictionary, accessing values in Dictionary. Traversing. Built-in- functions- len(), dict(), keys(), values(), items(), get(), update(), del(), clear(), pop(), popitems() etc. Programming examples.

Develop programs using Dictionary 1 Applying Lists: - Introduction: - values and accessing elements, List operations- concatenation, repetition and in operator. Traversing a List. Built-in List methods- len(), append(), insert(), index(), pop(), remove(), reverse() etc. Programming examples. Tuples: - Introduction: - creating Tuples, accessing values in Tuples, operations- concatenation, repetition and in operator. Built-in-Tuple functions- len(), max(), min() etc. Programming examples. Dictionaries:- Introduction: - creating Dictionary, accessing values in Dictionary. Traversing. Built-in- functions- len(), dict(), keys(), values(), items(), get(), update(), del(), clear(), pop(), popitems() etc. Programming examples. is an official topic in Module 3 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop programs using Dictionary 1 Applying Lists: - Introduction: - values and accessing elements, List operations- concatenation, repetition and in operator. Traversing a List. Built-in List methods- len(), append(), insert(), index(), pop(), remove(), reverse() etc. Programming examples. Tuples: - Introduction: - creating Tuples, accessing values in Tuples, operations- concatenation, repetition and in operator. Built-in-Tuple functions- len(), max(), min() etc. Programming examples. Dictionaries:- Introduction: - creating Dictionary, accessing values in Dictionary. Traversing. Built-in- functions- len(), dict(), keys(), values(), items(), get(), update(), del(), clear(), pop(), popitems() etc. Programming examples. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop programs using dictionary 1 applying lists: - introduction: - values and accessing elements, list operations- concatenation, repetition and in operator. traversing a list. built-in list methods- len(), append(), insert(), index(), pop(), remove(), reverse() etc. programming examples. tuples: - introduction: - creating tuples, accessing values in tuples, operations- concatenation, repetition and in operator. built-in-tuple functions- len(), max(), min() etc. programming examples. dictionaries:- introduction: - creating dictionary, accessing values in dictionary. traversing. built-in-functions- len(), dict(), keys(), values(), items(), get(), update(), del(), clear(), pop(), popitems() etc. programming examples. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop programs using Dictionary 1 Applying Lists: - Introduction: - values and accessing elements, List operations- concatenation, repetition and in operator. Traversing a List. Built-in List methods- len(), append(), insert(), index(), pop(), remove(), reverse() etc. Programming examples. Tuples: - Introduction: - creating Tuples, accessing values in Tuples, operations- concatenation, repetition and in operator. Built-in-Tuple functions- len(), max(), min() etc. Programming examples. Dictionaries:- Introduction: - creating Dictionary, accessing values in Dictionary. Traversing. Built-in- functions- len(), dict(), keys(), values(), items(), get(), update(), del(), clear(), pop(), popitems() etc. Programming examples. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Develop programs using Dictionary 1 Applying Lists: - Introduction: - values and accessing elements, List operations- concatenation, repetition and in operator. Traversing a List. Built-in List methods- len(), append(), insert(), index(), pop(), remove(), reverse() etc. Programming examples. Tuples: - Introduction: - creating Tuples, accessing values in Tuples, operations- concatenation, repetition and in operator. Built-in-Tuple functions- len(), max(), min() etc. Programming examples. Dictionaries:- Introduction: - creating Dictionary, accessing values in Dictionary. Traversing. Built-in- functions- len(), dict(), keys(), values(), items(), get(), update(), del(), clear(), pop(), popitems() etc. Programming examples. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
:
             problems.
            Explain the concept of Lists- accessing elements in
  M3.01     List and operations- concatenation, repetition, and in       1
Understanding
            operator
            Illustrate the concept of traversing a list and built-in-
  M3.02                                                                  1
Understanding
            List methods-len(), append(), insert() etc.

  M3.03     Develop programs using List                                  1
Applying
            Explain the concept of Tuples- creation, accessing
  M3.04     values in Tuples and operations- concatenation,              1
Understanding
            repetition, and in operator
            Summarize built-in-Tuple functions- len(), max(),
  M3.05                                                                  1
Understanding
            min() etc. – with examples.

  M3.06     Develop programs using Tuples                                1
Applying


            Explain how dictionaries are created, accessing
  M3.07                                                                  1
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

: Explain files, exceptions, and modules in Python for solving problems

This module is presented from the official syllabus scope for Problem Solving Using Python. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: : Explain files, exceptions, and modules in Python for solving problems
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

2 Understanding such as opening, reading, writing, closing and setting offsets in a file.

2 Understanding such as opening, reading, writing, closing and setting offsets in a file. is an official topic in Module 4 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding such as opening, reading, writing, closing and setting offsets in a file. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding such as opening, reading, writing, closing and setting offsets in a file. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding such as opening, reading, writing, closing and setting offsets in a file. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 2 Understanding such as opening, reading, writing, closing and setting offsets in a file. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the concept of pickle module with examples

Explain the concept of pickle module with examples is an official topic in Module 4 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the concept of pickle module with examples using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the concept of pickle module with examples should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the concept of pickle module with examples means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Explain the concept of pickle module with examples [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Develop programs for performing the operations on 2 Applying files Explain the concept of exceptions and built-in-

Develop programs for performing the operations on 2 Applying files Explain the concept of exceptions and built-in- is an official topic in Module 4 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop programs for performing the operations on 2 Applying files Explain the concept of exceptions and built-in- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop programs for performing the operations on 2 applying files explain the concept of exceptions and built-in- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop programs for performing the operations on 2 Applying files Explain the concept of exceptions and built-in- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Develop programs for performing the operations on 2 Applying files Explain the concept of exceptions and built-in- □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

1 Understanding exceptions

1 Understanding exceptions is an official topic in Module 4 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding exceptions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding exceptions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding exceptions means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 1 Understanding exceptions [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain how exceptions are handled with examples.

Explain how exceptions are handled with examples. is an official topic in Module 4 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain how exceptions are handled with examples. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain how exceptions are handled with examples. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain how exceptions are handled with examples. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain how exceptions are handled with examples. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Develop programs for handling exceptions. 2 Applying Files: - Introduction: - Types of files- Text, binary and CSV files. Opening, closing, reading and writing in a text binary and CSV files. Setting offsets in a file. Pickle module- load() and dump(). Programming examples Exceptions: - Exceptions, built-in-exceptions, handling exceptions – Programming examples. Text / Reference T/R Book Title/Author T1 Balagurusamy E, Introduction to Computing and Problem Solving Using Python. R1 Lambert K.A, Fundamentals of Python-First Programs. Cengage Learning India 2015. R2 Perkovic .L., Introduction to Computing Using Python, 2nd Edition R3 Zelle. J., Python Programming: An Introduction to Computer Science. R4 Allen Downey, Jeffrey Elkner, Chris Meyers., How to think like a computer scientist : learning with Python, 1st Edition Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com/python/index.htm 2 https://www.javatpoint.com/python-tutorial 3 https://www.programiz.com/python-programming 4 https://www.w3schools.com/python/default.asp

Develop programs for handling exceptions. 2 Applying Files: - Introduction: - Types of files- Text, binary and CSV files. Opening, closing, reading and writing in a text binary and CSV files. Setting offsets in a file. Pickle module- load() and dump(). Programming examples Exceptions: - Exceptions, built-in-exceptions, handling exceptions – Programming examples. Text / Reference T/R Book Title/Author T1 Balagurusamy E, Introduction to Computing and Problem Solving Using Python. R1 Lambert K.A, Fundamentals of Python-First Programs. Cengage Learning India 2015. R2 Perkovic .L., Introduction to Computing Using Python, 2nd Edition R3 Zelle. J., Python Programming: An Introduction to Computer Science. R4 Allen Downey, Jeffrey Elkner, Chris Meyers., How to think like a computer scientist : learning with Python, 1st Edition Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com/python/index.htm 2 https://www.javatpoint.com/python-tutorial 3 https://www.programiz.com/python-programming 4 https://www.w3schools.com/python/default.asp is an official topic in Module 4 of Problem Solving Using Python. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop programs for handling exceptions. 2 Applying Files: - Introduction: - Types of files- Text, binary and CSV files. Opening, closing, reading and writing in a text binary and CSV files. Setting offsets in a file. Pickle module- load() and dump(). Programming examples Exceptions: - Exceptions, built-in-exceptions, handling exceptions – Programming examples. Text / Reference T/R Book Title/Author T1 Balagurusamy E, Introduction to Computing and Problem Solving Using Python. R1 Lambert K.A, Fundamentals of Python-First Programs. Cengage Learning India 2015. R2 Perkovic .L., Introduction to Computing Using Python, 2nd Edition R3 Zelle. J., Python Programming: An Introduction to Computer Science. R4 Allen Downey, Jeffrey Elkner, Chris Meyers., How to think like a computer scientist : learning with Python, 1st Edition Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com/python/index.htm 2 https://www.javatpoint.com/python-tutorial 3 https://www.programiz.com/python-programming 4 https://www.w3schools.com/python/default.asp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop programs for handling exceptions. 2 applying files: - introduction: - types of files- text, binary and csv files. opening, closing, reading and writing in a text binary and csv files. setting offsets in a file. pickle module- load() and dump(). programming examples exceptions: - exceptions, built-in-exceptions, handling exceptions – programming examples. text / reference t/r book title/author t1 balagurusamy e, introduction to computing and problem solving using python. r1 lambert k.a, fundamentals of python-first programs. cengage learning india 2015. r2 perkovic .l., introduction to computing using python, 2nd edition r3 zelle. j., python programming: an introduction to computer science. r4 allen downey, jeffrey elkner, chris meyers., how to think like a computer scientist : learning with python, 1st edition online resources sl.no website link 1 https://www.tutorialspoint.com/python/index.htm 2 https://www.javatpoint.com/python-tutorial 3 https://www.programiz.com/python-programming 4 https://www.w3schools.com/python/default.asp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop programs for handling exceptions. 2 Applying Files: - Introduction: - Types of files- Text, binary and CSV files. Opening, closing, reading and writing in a text binary and CSV files.

Aspect	What to prepare
	Setting offsets in a file. Pickle module- load() and dump(). Programming examples Exceptions: - Exceptions, built-in-exceptions, handling exceptions – Programming examples. Text / Reference T/R Book Title/Author T1 Balagurusamy E, Introduction to Computing and Problem Solving Using Python. R1 Lambert K.A, Fundamentals of Python-First Programs. Cengage Learning India 2015. R2 Perkovic .L., Introduction to Computing Using Python, 2nd Edition R3 Zelle. J., Python Programming: An Introduction to Computer Science. R4 Allen Downey, Jeffrey Elkner, Chris Meyers., How to think like a computer scientist : learning with Python, 1st Edition Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com/python/index.htm 2 https://www.javatpoint.com/python-tutorial 3 https://www.programiz.com/python-programming 4 https://www.w3schools.com/python/default.asp means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Develop programs for handling exceptions. 2 Applying Files: - Introduction: - Types of files- Text, binary and CSV files. Opening, closing, reading and writing in a text binary and CSV files. Setting offsets in a file. Pickle module- load() and dump(). Programming examples Exceptions: - Exceptions, built-in-exceptions, handling exceptions – Programming examples. Text / Reference T/R Book Title/Author T1 Balagurusamy E, Introduction to Computing and Problem Solving Using Python. R1 Lambert K.A, Fundamentals of Python-First Programs. Cengage Learning India 2015. R2 Perkovic .L., Introduction to Computing Using Python, 2nd Edition R3 Zelle. J., Python Programming: An Introduction to Computer Science. R4 Allen Downey, Jeffrey Elkner, Chris Meyers., How to think like a computer scientist : learning with Python, 1st Edition Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com/python/index.htm 2 https://www.javatpoint.com/python-tutorial 3 https://www.programiz.com/python-programming 4 https://www.w3schools.com/python/default.asp □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
:       Explain files, exceptions, and modules in Python for solving problems
            Explain the concept of files and types of files.
            Perform the operations on text, binary and csv files
  M4.01                                                                  2
Understanding
            such as opening, reading, writing, closing and
            setting offsets in a file.
  M4.02     Explain the concept of pickle module with examples           1
Understanding
  M4.03     Develop programs for performing the operations on            2
Applying
            files
            Explain the concept of exceptions and built-in-
  M4.04                                                              1
Understanding
            exceptions
  M4.05     Explain how exceptions are handled with examples.        1
Understanding
  M4.06     Develop programs for handling exceptions.                2
Applying
            Series Test – II                                         1
Contents:
Files: -
Introduction: - Types of files- Text, binary and CSV files. Opening, closing,
reading and
writing in a text binary and CSV files. Setting offsets in a file. Pickle module-
load() and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 1 Understanding. algorithm, flow chart, and pseudo code. Illustrate the basic concept of a Python- identifiers,. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding. algorithm, flow chart, and pseudo code. Illustrate the basic concept of a Python- identifiers,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding keywords, variables and data types. Explain operators and expressions- Expressions, Arithmetic, comparison, assignment, logical, bitwise,. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding keywords, variables and data types. Explain operators and expressions- Expressions, Arithmetic, comparison, assignment, logical, bitwise,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding membership and identity operators, precedence of operators, associativity and input output functions. Explain Control statements- if elif else statement,. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding membership and identity operators, precedence of operators, associativity and input output functions. Explain Control statements- if elif else statement,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain while statement, for statement, break and continue 3 Understanding statements.. (3 marks)

Answer: Begin with the standard definition or identity of *while statement, for statement, break and continue 3 Understanding statements.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain built-in-functions, type conversion, type coercion, 2 Understanding. mathematical functions Explain the concept of user defined functions-. (3 marks)

Answer: Begin with the standard definition or identity of *built-in-functions, type conversion, type coercion, 2 Understanding. mathematical functions Explain the concept of user defined functions-*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain function definition, function calling, function 3 Understanding returning values Illustrate the difference between actual and formal. (3 marks)

Answer: Begin with the standard definition or identity of *function definition, function calling, function 3 Understanding returning values Illustrate the difference between actual and formal*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain parameters. Types of formal arguments- required, 2 Understanding. default and keyword arguments. Explain the concept of scope of variable-local and. (3 marks)

Answer: Begin with the standard definition or identity of *parameters. Types of formal arguments- required, 2 Understanding. default and keyword arguments. Explain the* *concept of scope of variable-local and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding. global scope and recursive function.. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding. global scope and recursive function.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain List and operations- concatenation, repetition, and in 1 Understanding operator Illustrate the concept of traversing a list and built-in-. (3 marks)

Answer: Begin with the standard definition or identity of *List and operations- concatenation, repetition, and in 1 Understanding operator Illustrate the concept of traversing a list and built-in-*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding List methods-len(), append(), insert() etc.. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding List methods- len(), append(), insert() etc.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Develop programs using List 1 Applying Explain the concept of Tuples- creation, accessing. (3 marks)

Answer: Begin with the standard definition or identity of *Develop programs using List 1 Applying Explain the concept of Tuples- creation, accessing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain values in Tuples and operations- concatenation, 1 Understanding repetition, and in operator Summarize built-in-Tuple functions- len(), max(),. (3 marks)

Answer: Begin with the standard definition or identity of *values in Tuples and operations- concatenation, 1 Understanding repetition, and in operator Summarize built-in-Tuple functions- len(), max(),*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 2 Understanding such as opening, reading, writing, closing and setting offsets in a file.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding such as opening, reading, writing, closing and setting offsets in a file.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition, ▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯.

Define or explain Explain the concept of pickle module with examples. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the concept of pickle module with examples*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition, ▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯.

Define or explain Develop programs for performing the operations on 2 Applying files Explain the concept of exceptions and built-in-. (3 marks)

Answer: Begin with the standard definition or identity of *Develop programs for performing the operations on 2 Applying files Explain the concept of exceptions and built-in-*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition, ▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯.

Define or explain 1 Understanding exceptions. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding exceptions*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Understanding. algorithm, flow chart, and pseudo code. Illustrate the basic concept of a Python- identifiers,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **built-in-functions, type conversion, type coercion, 2 Understanding. mathematical functions Explain the concept of user defined functions-**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **List and operations- concatenation, repetition, and in 1 Understanding operator Illustrate the concept of traversing a list and built-in-**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **2 Understanding such as opening, reading, writing, closing and setting offsets in a file.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://www.tutorialspoint.com/python/index.htm https://www.javatpoint.com/python-tutorial https://www.programiz.com/python-programming https://www.w3schools.com/python/default.asp

IN-PAGE SEARCH

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